



DADS 5001

DATA ANALYTICS AND DATA SCIENCE TOOLS AND PROGRAMMING

INTRODUCTION & PANDAS 1

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OUTLINE

- ✕ Course's Introduction
- ✕ Mini-project
- ✕ Package installation
- ✕ Exploratory Data Analysis
- ✕ List vs. Pandas vs. NumPy
- ✕ Pandas



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COURSE'S INTRODUCTION

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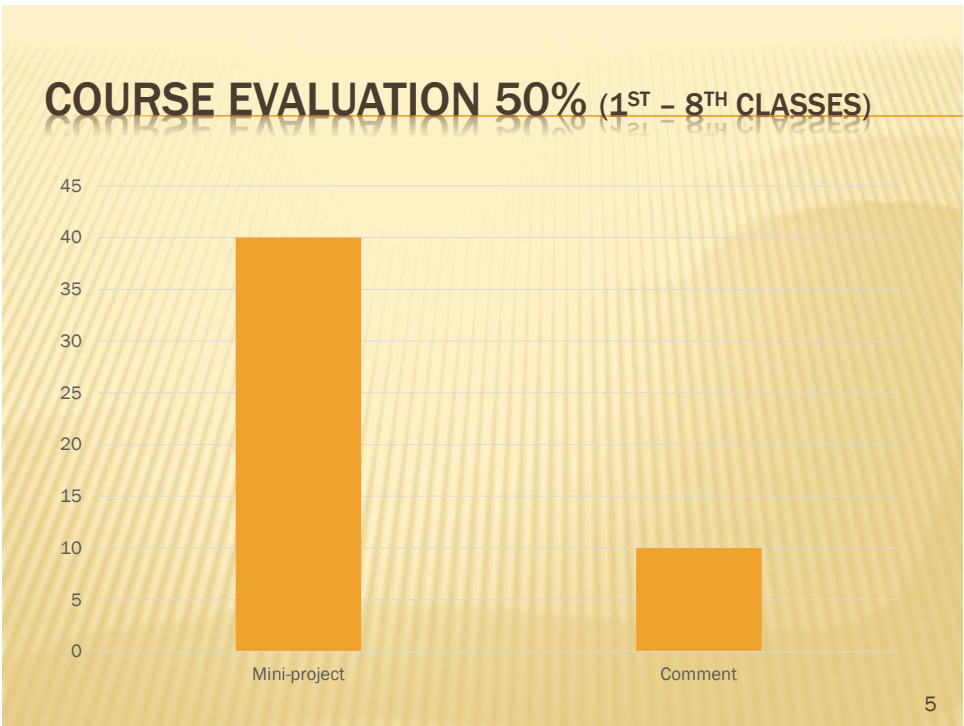
MICROSOFT TEAMS

✕ (1/2569) DADS5001

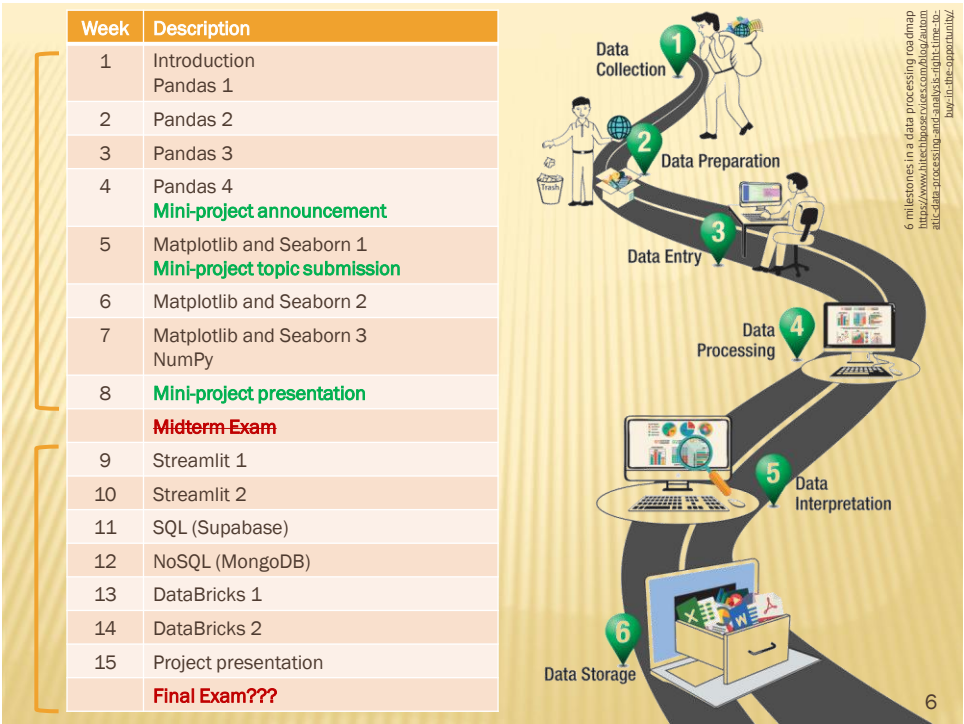
- + **Official announcement:** General channel
- + **Class material (all in English):** General channel > Shared (Files) tab > Class Materials folder
- + **Syllabus:** General channel > Shared (Files) tab > Class Materials folder > COURSE_SYLLABUS.pdf
- + **Latest class schedule:** General channel > Shared (Files) tab > Class Materials folder > CLASS_SCHEDULE.docx

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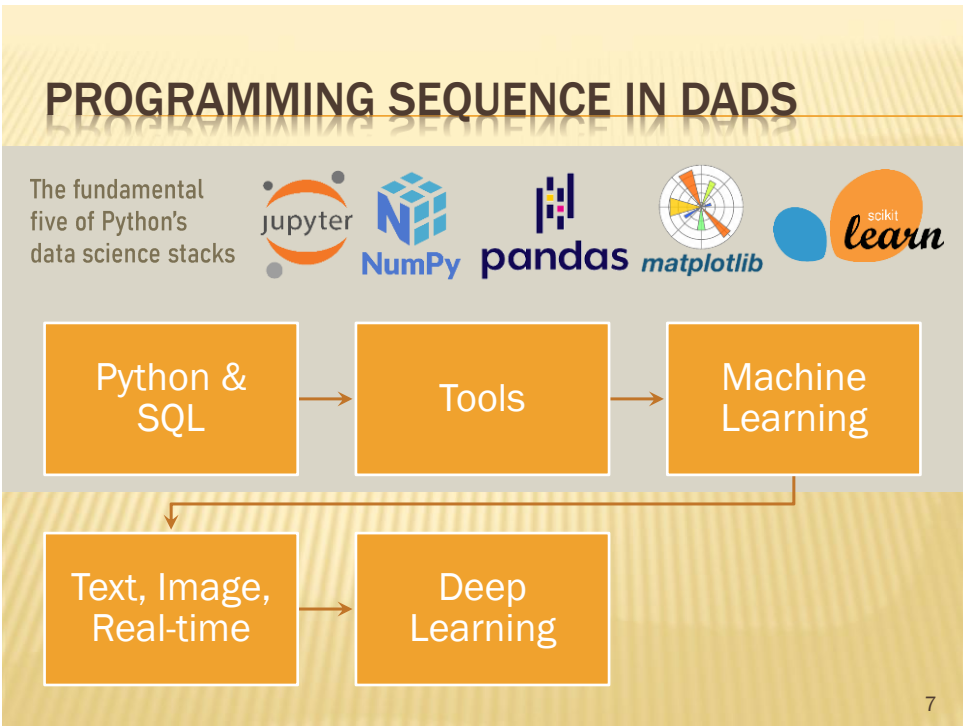
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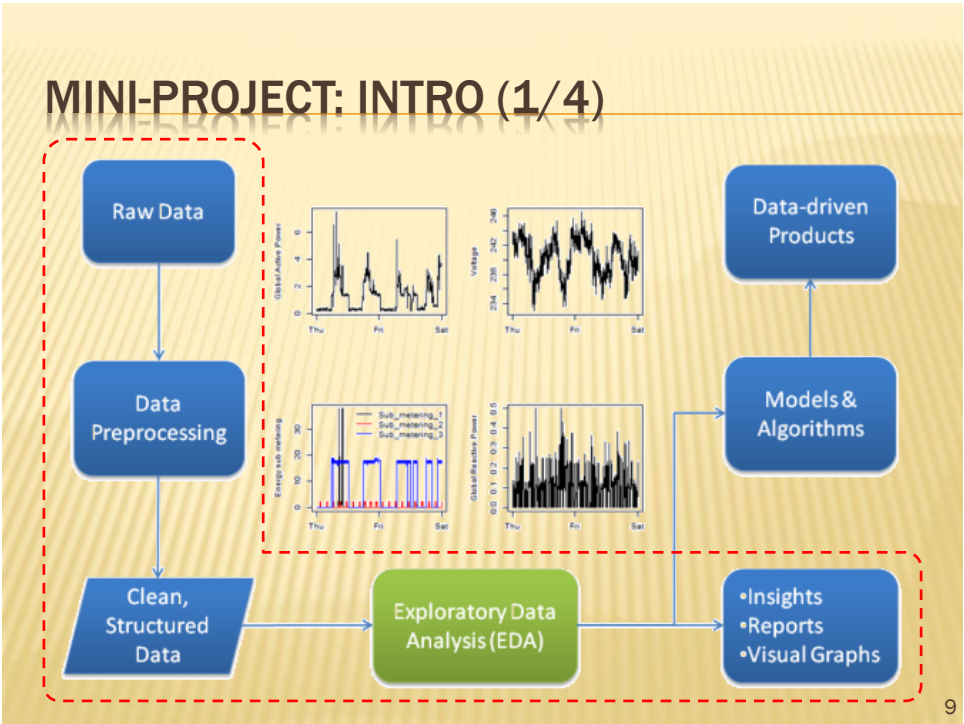
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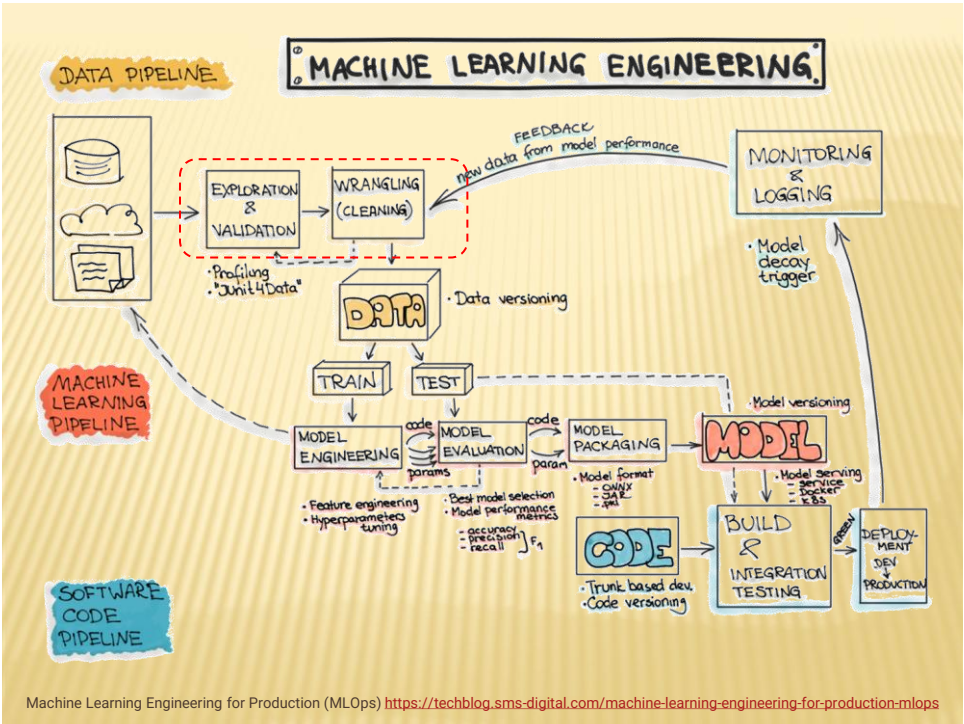
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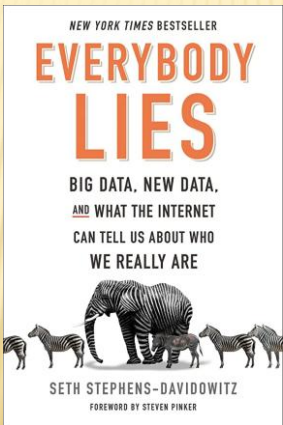


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MINI-PROJECT: INTRO (3/4)

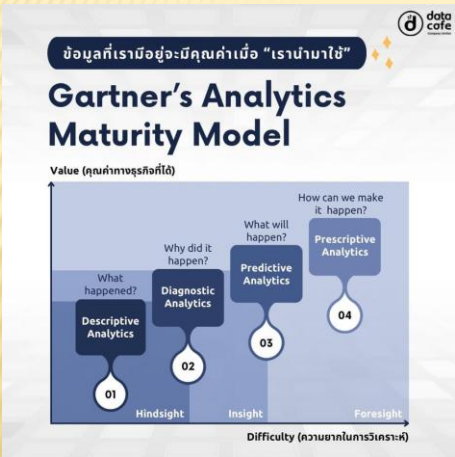
“Most important, to squeeze insights out of Big Data, *you have to ask the right questions.*”



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MINI-PROJECT: INTRO (4/4)



มีการตั้งสมมติฐาน และการตีโจทย์ปัญหาต่างๆ ในการวิเคราะห์ข้อมูลยัง任重道远

สอบถาม: ตอนนี้นำสิ่งที่เรียน Data มาแก้ปัญหาสำคัญแล้วคือ การตั้งสมมติฐานในการวิเคราะห์ข้อมูล, การตีโจทย์ปัญหาต่างๆ ต้องศึกษาเพิ่มเติมอย่างไรดีคะ

มาแชร์กันครับ :)

DATA LAZY

<https://www.facebook.com/100057174082208/posts/pfbid022YgWGJAUUTXkPXN9gho825NkL9NVc7qNnd5dhTJZg8QnBIT6ujy1K7xzNZ82bEHn/?mibextid=cr9u03>, <https://www.facebook.com/d4biz/photos/a.3233628596865997/3229272480634942>

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MINI-PROJECT: HOW TO

- 1. Having a good question is a good start, so as an initial assumption.
- 2. Check data availability/accessibility (realistic and unprocessed data, not the already-prepped neat dataset with abundant tutorials/examples)
- 3. Prove or disprove your assumption with data analytics and data science
 - + **Programming tools and (static) visualization must be included significantly.**
 - + The data source may be shared among some students but what comes afterward (questioning, cleaning, processing, EDA, visualizing) must be done individually.
- 4. Submit as a public GitHub link (one link per one group), including codes and development journey



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MINI-PROJECT: EXAMPLES (1/3)

- ✕ ส่อง Mega trend "EV Car" <https://github.com/ssorawits/Project.git>
- ✕ การวิเคราะห์การเข้ารับบริการในฐานผู้ป่วยในของผู้ป่วยสิทธิหลักประกันสุขภาพแห่งชาติ <https://github.com/ChotepipatC/DADS5001>
- ✕ Global Health & Pharmaceutical Spending: Aging, Disease Burden, and Investment Insights <https://github.com/polyst08/pandasproject-healthexpenditure>
- ✕ ศึกษาวิเคราะห์เหตุการณ์และผลกระทบในแผ่นดินไหวของประเทศไทย <https://github.com/tontantip/DADS5001-Project-Earthquake/>
- ✕ อุบัติเหตุคือเธอไม่รัก <https://github.com/techasit239/Dads-5001-Accident-Is-You-Dont-Love>
- ✕ Stock screening using financial ratios <https://github.com/suparuek2405/Stock-screening-with-python/tree/main>
- ✕ อิทธิพลของ Stephen Curry กับการบาสเกตบอล NBA ที่เปลี่ยนไป <https://github.com/ThiwakornC/Spotter-NBA/tree/main>
- ✕ Life expectancy analysis https://github.com/bwinna/mini_project_World-why-wealth
- ✕ ผลกระทบจากเศรษฐกิจทำให้อัตราเด็กเกิดใหม่ลดลงหรือไม่ https://github.com/pnithida/6420422004_MiniProject
- ✕ PM2.5 in Thailand(Y2019-Y2021) <https://github.com/Porrakij/PM2.5-Data-Analysis>
- ✕ สำราจระเบิดคันคนไทย รายได้ และ รายจ่าย เติลยของคร้าเรือน https://github.com/koraweeep/DADS5001_Mini-Project
- ✕ จังหวัดเฝ้าระวัง เพื่อป้องกันและลดจำนวนผู้เสียชีวิตจากอุบัติเหตุทางถนน https://github.com/ploychenya/DADS5001_mini_project
- ✕ Big Data กับโลกอสังหาริมทรัพย์ <https://github.com/Hakulani/miniprojectDADS5001>

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MINI-PROJECT: EXAMPLES (2/3)

- ✖ เลี้ยงลูกไม่ไหว เลี้ยงสัตว์ก็ไม่รอด https://github.com/Keeta-Kimila/Beyond_Babies_Pet_Ownership_as_Thailand_Emerging_Customer_Market
- ✖ ข้อมูลมูลค่าการนำเข้าสินค้าของประเทศไทยจากทั่วโลกในช่วงปี 2002 – 2022 https://github.com/nacknatthawit/DADS5001_6420412006
- ✖ Thailand_Accident_Jan-Jun2022 https://github.com/waewma/Thailand_Accident_Jan-Jun2022
- ✖ ยอดการค้นหา Google trend ด้วย keyword "Bitkub" จะส่งผลดีต่อบริการของตัวบริษัทหรือไม่ https://github.com/kimteespk/DADS5001_Miniproject_bitkub_googletrend_6510412011
- ✖ Vending machine analysis <https://github.com/kikkalo/6420412010>
- ✖ Thailand Labour Demand Trend Y2015-2020 <https://github.com/Sujitra17/Mini-Project>
- ✖ Are Thai healthcare related stock still captivating? <https://github.com/mmaiip/project-5001>
- ✖ Global warming https://github.com/Hellper1/DADS_5001_miniproject
- ✖ Video Game Sales <https://github.com/MeenWhile/Tools-Mini-Project>
- ✖ World-Population Analysis <https://github.com/o-joe-v/World-Population>
- ✖ Thailand and World Happiness https://github.com/HikariJadeEmpire/TH_WORLD_Happiness_Project
- ✖ สถานการณ์อุตสาหกรรมรถยนต์ไทย <https://github.com/crispyporkwithholylbasil/5001-DADS-miniproject>
- ✖ Hollywood Insight https://github.com/y-lims/DADS5001_Hollywood_Insight
- ✖ ค่าไฟแพงมาจากอะไร? สถานการณ์พลังงานไฟฟ้าในประเทศไทย <https://github.com/kkengg/Pikachu-Project>

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MINI-PROJECT: EXAMPLES (3/3)

- ✖ From 2026 until now: (wide adoption of Agentic AI)
 - + การวิเคราะห์กลยุทธ์เพื่อเจาะลึกประสิทธิภาพการจัดการ HIV ของประเทศไทย (2021-2025): เมื่อ “ปริมาณ” อาจไม่ใช่คำตอบของ “ประสิทธิภาพ” https://github.com/sittidetw/dads5001_hiv_strategic_analysis
 - + บทวิเคราะห์ผลการศึกษา: โครงสร้างภาษีของประเทศไทย - ใครคือผู้แบกรับภาระที่แท้จริง? https://github.com/punnarujc/dads5001_midterm_project
 - + การวิเคราะห์ผ่านเลนส์ของระบบขนส่งสาธารณะ ในการทำความเข้าใจว่ามหานคร ASEAN กำลังขยับตัวไปในทิศทางใด https://github.com/Kaitokidbua/ASEAN_Transport/blob/main/README.md
 - + การวิเคราะห์ผลการเลือกตั้งปี 2569 https://github.com/Gorszet/DADS5001_Election_2569

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MINI-PROJECT: EXAMPLES (3/3)

✗ From 2026 until now: (wide adoption of Agentic AI)

What changed	Suggestion
Polished language ✓	-
Overly verbose text ✗	Keep sentences short, punchy, and clear
Unsubstantiated conclusions ✗✗	Ensure every insight is directly supported by the source data or the proposed analysis
Obvious AI conversational traces ✗✗✗	“ลองสังเกตที่ขนาดบรีฟที่ครับ”

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ARE DATA JOBS THE NEXT TO FALL?

✗ Working with AI: Measuring the Occupational Implications of Generative AI

Microsoft 22JUL2025:
<https://arxiv.org/abs/2507.07935>



Table 3: Top 40 occupations with highest AI applicability score.

Job Title (Abbrev.)	Coverage	Cmpltn.	Scope	Score	Employment
Interpreters and Translators	0.98	0.88	0.57	0.49	51,560
Historians	0.91	0.85	0.56	0.48	3,040
Passenger Attendants	0.80	0.88	0.62	0.47	20,190
Sales Representatives of Services	0.84	0.90	0.57	0.46	1,142,020
Writers and Authors	0.85	0.84	0.60	0.45	49,450
Customer Service Representatives	0.72	0.90	0.59	0.44	2,858,710
CNC Tool Programmers	0.90	0.87	0.53	0.44	28,030
Telephone Operators	0.80	0.86	0.57	0.42	4,600
Ticket Agents and Travel Clerks	0.71	0.90	0.56	0.41	119,270
Broadcast Announcers and Radio DJs	0.74	0.84	0.60	0.41	25,070
Brokerage Clerks	0.74	0.89	0.57	0.41	48,060
Farm and Home Management Educators	0.77	0.91	0.55	0.41	8,110
Telemarketers	0.66	0.89	0.60	0.40	81,580
Concierges	0.70	0.88	0.56	0.40	41,020
Political Scientists	0.77	0.87	0.53	0.39	5,580
News Analysts, Reporters, Journalists	0.81	0.81	0.56	0.39	45,020
Mathematicians	0.91	0.74	0.54	0.39	2,220
Technical Writers	0.83	0.82	0.54	0.38	47,970
Proofreaders and Copy Markers	0.91	0.86	0.49	0.38	5,490
Hosts and Hostesses	0.60	0.90	0.57	0.37	425,020
Editors	0.78	0.82	0.54	0.37	95,700
Business Teachers, Postsecondary	0.70	0.90	0.52	0.37	82,980
Public Relations Specialists	0.63	0.90	0.60	0.36	275,550
Demonstrators and Product Promoters	0.64	0.88	0.53	0.36	50,790
Advertising Sales Agents	0.66	0.90	0.53	0.36	108,100
New Accounts Clerks	0.72	0.87	0.51	0.36	41,180
Statistical Assistants	0.85	0.84	0.49	0.36	7,200
Counter and Rental Clerks	0.62	0.90	0.52	0.36	390,300
Data Scientists	0.77	0.86	0.51	0.36	192,710
Personal Financial Advisors	0.69	0.88	0.52	0.35	272,190
Archivists	0.66	0.88	0.49	0.35	7,150
Economics Teachers, Postsecondary	0.68	0.90	0.51	0.35	12,210
Web Developers	0.73	0.86	0.51	0.35	85,350
Management Analysts	0.68	0.90	0.54	0.35	838,140
Geographers	0.77	0.83	0.48	0.35	1,460
Models	0.64	0.89	0.53	0.35	3,090
Market Research Analysts	0.71	0.90	0.52	0.35	846,370
Public Safety Telecommunicators	0.66	0.88	0.53	0.35	97,820
Switchboard Operators	0.68	0.86	0.52	0.35	43,830
Library Science Teachers, Postsecondary	0.65	0.90	0.51	0.34	4,220

Note: Metrics reported as mean of user goal and AI action score.

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PACKAGE INSTALLATION

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PIP INSTALL VS. CONDA INSTALL

PACKAGE AND ENVIRONMENT MANAGERS



pip: install packages within any environment.

- Limited to Python-only packages and dependencies.
- No effort is made to ensure that the dependencies of all packages are fulfilled simultaneously.
- Requires separate tools for environment management.
- Can mess up the system Python (admin w/o venv).
- Lot of available packages in PyPI.

conda: install packages within conda environments only.

- Manages system-level packages and dependencies (C/C++, CUDA) not just Python.
- Uses a robust SAT (satisfiability) solver for conflict prevention.
- Easy to maintain environments.
- Fewer overall packages than PyPI.
- Only free for individuals, small organizations, and any size of educational entities.

<https://www.anaconda.com/blog/understanding-conda-and-pip> 54

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REQUIRED LIBRARIES

```
conda install jupyter
```

```
conda install pandas
```

```
conda install numpy
```

```
conda install matplotlib
```

```
conda install seaborn
```

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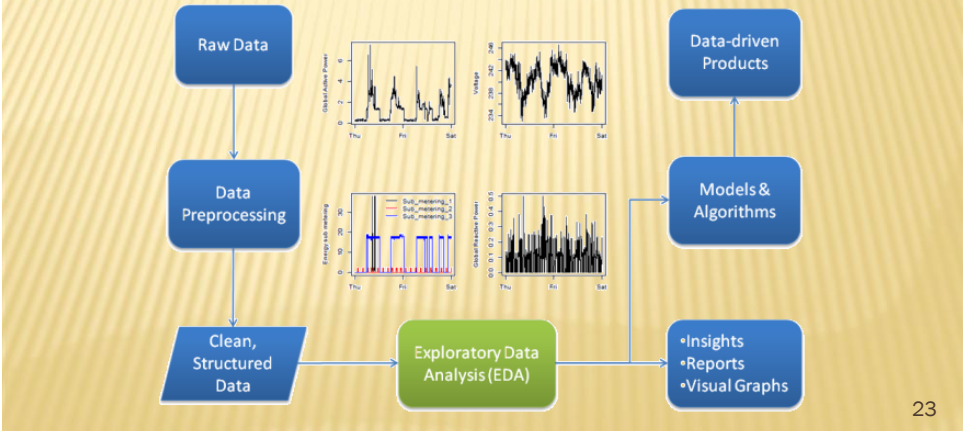
EXPLORATORY DATA ANALYSIS

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EXPLORATORY DATA ANALYSIS (1/4)

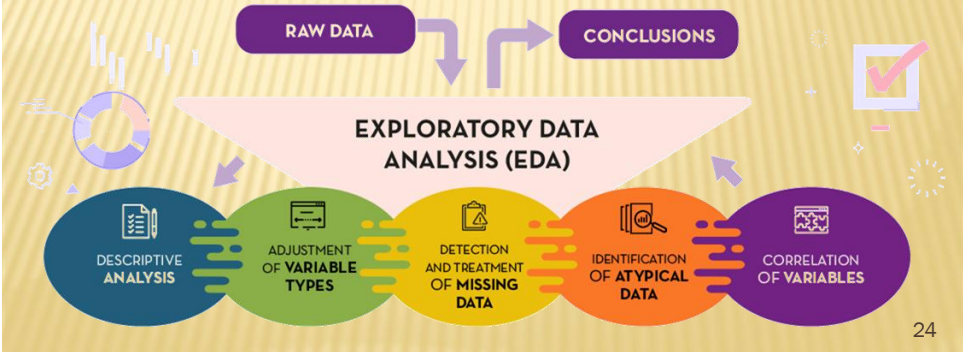
- ✖ Originally developed by American mathematician John Tukey in the 1970s, **EDA** techniques continue to be a widely used method in the data discovery process today.



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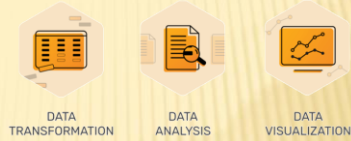
EXPLORATORY DATA ANALYSIS (2/4)

- ✖ **EDA** is used by data analysts/scientists to analyze and investigate data sets and summarize their main characteristics, often employing data visualization methods.
- ✖ The main purpose of **EDA** is to help look at data before making any assumptions.



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- ✗ **EDA** is an iterative process of



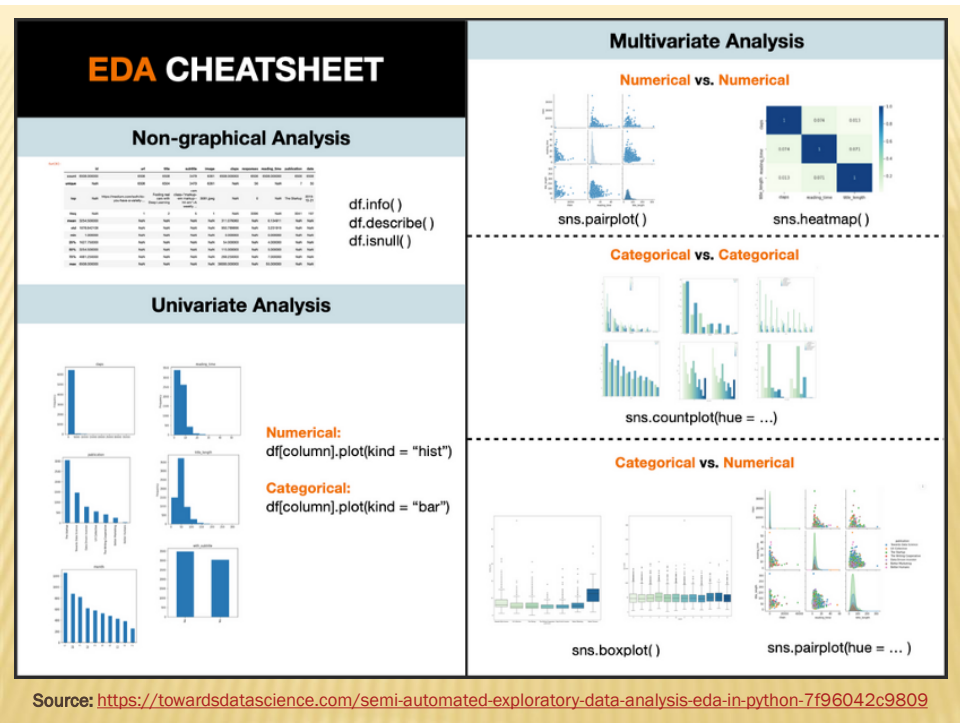
- ✗ What for?

- + Provide a better understanding of data set patterns, variables, and the relationships between them
- + Help identify obvious errors and detect outliers or anomalous events
- + See what data can reveal beyond the formal modeling or hypothesis testing task
- + Help determine if the statistical techniques we are considering for data analysis are appropriate
- + Ensure the results we produce are valid and applicable to any desired business outcomes and goals

- ✖ Once **EDA** is complete and insights are drawn, its features can then be used for more sophisticated data analysis or modeling, including machine learning and deep learning.

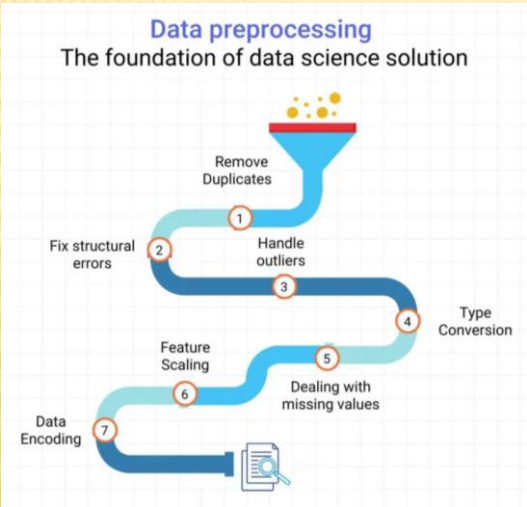
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DATA PREPROCESSING (1/2)



✗ A common mistake that many novice data scientists make is that they skip through the data wrangling stage and dive right into the model-building phase, which in turn generates a poor-performing machine learning model.

21NOV2022: <https://datasciencedojo.com/blog/data-preprocessing-data-science-solution/>

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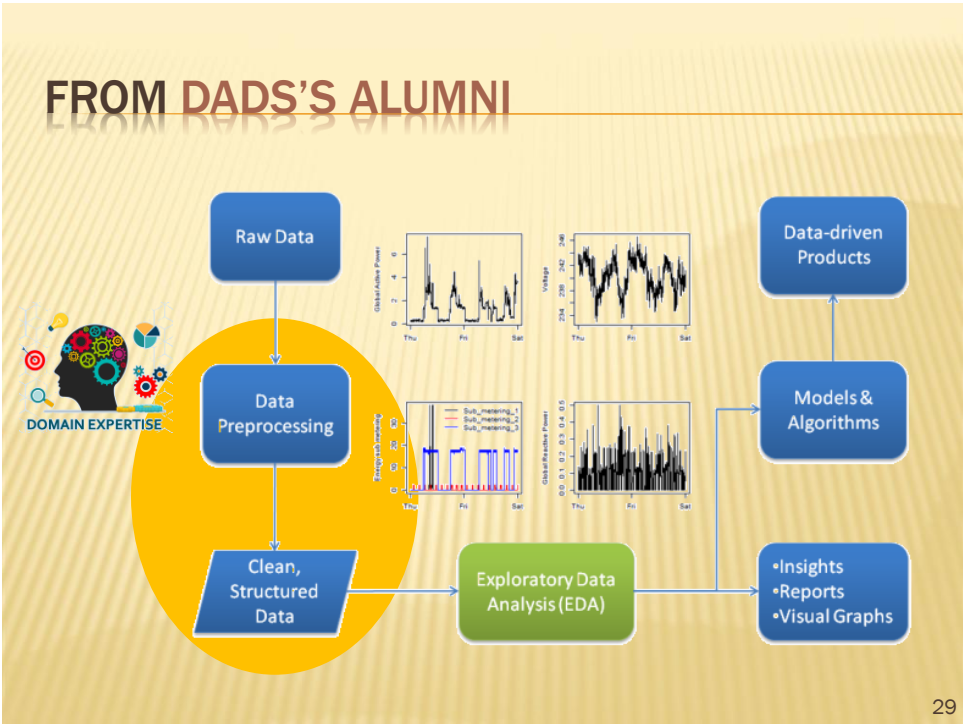
DATA PREPROCESSING (2/2)

- 1. Remove duplicates
 - + Repeated entries will lead to the model overfitting.
 - + Be careful when there are too many duplicate entries.
- 2. Fix structural errors
 - + Structural errors in a dataset refer to the entries that either have typos or inconsistent spellings.
 - + Check all unique values and their corresponding occurrence
- 3. Detect and handle outliers
 - + Outlier is any value in a dataset that drastically deviates from the rest of the data points. It can mess up our machine-learning model if not taken care of.
 - + Use the describe function on columns, visualize outliers with box plots, compute z-score on columns (99.7% of the data points within the range of -3 and +3 scores), etc.
- 4. Type conversion
 - + Do type conversion when certain columns are not of valid data type (e.g., the object data type)
- 5. Dealing with missing values
 - + Often, data set contains numerous missing values, which can be a problem. For example, it can play a role in the development of a biased estimator, or it can decrease the representativeness of the sample under consideration.
 - + Drop rows with missing values, impute the missing values with central tendencies (e.g., mean, median, mode), forward/backward fill
- 6. Feature scaling
- 7. Data encoding

Credit (21NOV2022): <https://datasciencedojo.com/blog/data-preprocessing-data-science-solution/>

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PYTHON'S LIST VS. PANDAS VS. NUMPY

PYTHON'S LIST

✖ <https://wiki.python.org/moin/TimeComplexity>

list

The Average Case assumes parameters generated uniformly at random.

Internally, a list is represented as an array; the largest costs come from growing beyond the current allocation size (because everything must move), or from inserting or deleting somewhere near the beginning (because everything after that must move). If you need to add/remove at both ends, consider using a `collections.deque` instead.

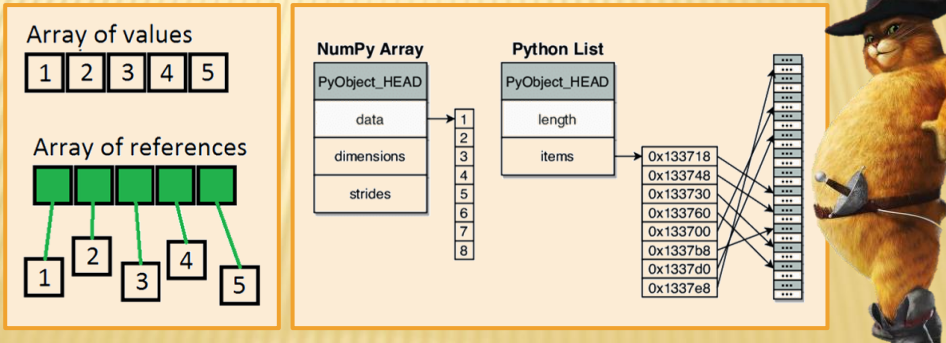


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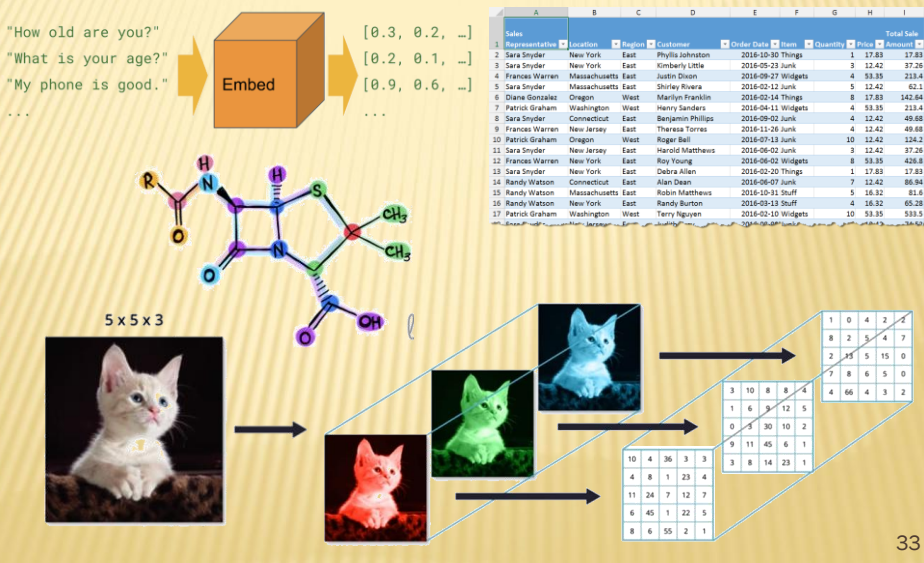
ARRAY VS. PYTHON'S LIST

- ✖ In other programming languages, **Array** refers to the static memory allocation for holding a sequence of homogeneous data. **Array** is faster.
- ✖ Because Python's **List** stores references to data, a **List** can (seemingly) hold heterogeneous data. **List** is slower.



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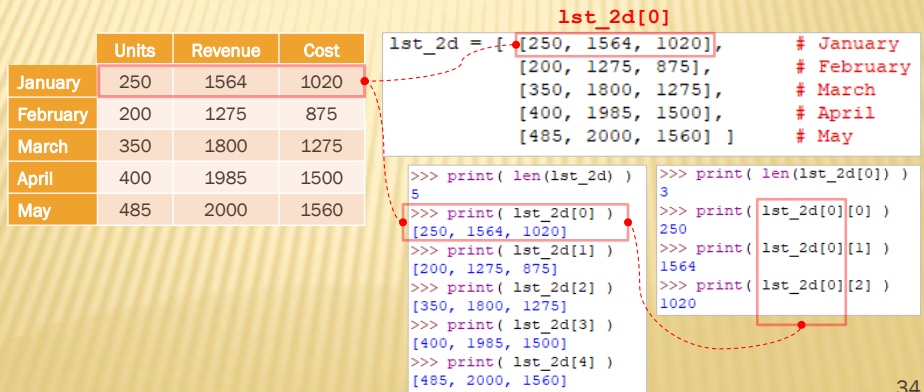
REAL-WORLD DATA IS NOT 1D



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PYTHON'S LIST FOR MULTI-DIM DATA

- ✗ Possible but not good
 - + Slow for large data
 - + Inconvenient access/iteration/calculation on multi-dim data



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PANDAS (2008) VS. EXCEL (1985)

- ✗ Too much data slows down Excel.
- ✗ Excel is limited to 1,048,576 rows by 16,384 columns per worksheet. Exceeding the limit causes data loss.
- ✗ **Pandas** is one of the most popularly used Python modules for data analysis and manipulation. It's said to be a data analyst's best friend.
- ✗ No limits to the size of data, powerful data transformation, allow automation and extended file formats
- ✗ **Pandas** is good and intuitive for tabular data (e.g., csv, excel, sql) and it includes time series functionalities.



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PANDAS (2008) VS. NUMPY (2005)

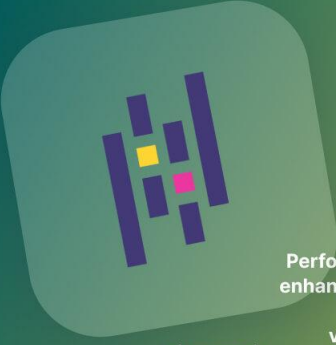
- ✗ **Pandas** is built on top of two core Python libraries—matplotlib for data visualization and **NumPy** for mathematical operations.
- ✗ **Pandas** acts as a wrapper over these two libraries, allowing us to access many of matplotlib's and **NumPy**'s methods with less code.
- ✗ **Pandas** for data analysts. **NumPy** for data scientists
- ✗ **Pandas** is popularly used for data analysis and manipulation. **NumPy** is primarily used for numerical calculations.
- ✗ **Pandas** is more natural for database-like data (e.g., csv, excel, sql). **NumPy** is more natural for numeric processing of data (e.g., signals, images).
- ✗ Scikit-learn was originally developed to work well with **NumPy** array. It's recommended to use **NumPy** array with Scikit-learn due to mature data handling.

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PANDAS 2.0 (1/4)

What's new in 2.0.0 (April 3, 2023): <https://pandas.pydata.org/docs/whatsnew/v2.0.0.html>



Pandas 2.0

13+ years of Pandas and its ecosystem (Arrow, Polars, DuckDB)

Performance enhancements

Init commit

v0.5

Big improvements and enhancements

v1.0

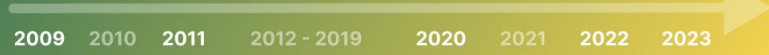
Python 3.8

Arrow support

More Arrow

v1.5

v2.0



2009 2010 2011 2012 - 2019 2020 2021 2022 2023

Credit: <https://airbyte.com/blog/pandas-2-0-ecosystem-arrow-polars-duckdb>

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PANDAS 2.0 (2/4)

What's new in 2.0.0 (April 3, 2023): <https://pandas.pydata.org/docs/whatsnew/v2.0.0.html>

✖ Pandas 2.0 is faster and more memory efficient with the PyArrow backend (Apache Arrow).

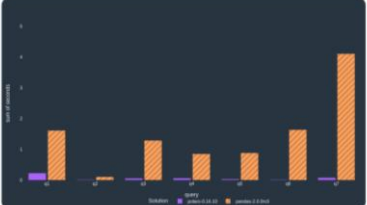


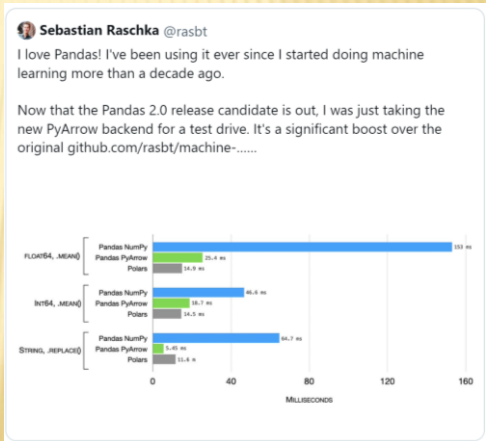
Ritchie Vink (@ritchie46@fosstodon.org)
@RitchieVink Follow

Pandas 2.0 vs Polars 0.16.10

I see people taking the wrong conclusions from this microbenchmark. Here are the results of pandas with arrow backed datatypes vs polars on TPCH queries.

Micro benchmarks don't show the whole query runtime. Code here: github.com/pola-rs/tpch/p...

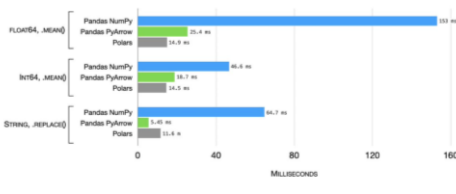




Sebastian Raschka @rasbt

I love Pandas! I've been using it ever since I started doing machine learning more than a decade ago.

Now that the Pandas 2.0 release candidate is out, I was just taking the new PyArrow backend for a test drive. It's a significant boost over the original github.com/rasbt/machine-...



Credit: <https://twitter.com/RitchieVink/status/1632334005264580608>

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PPT template, images and decorating graphics from www.google.com unless otherwise specified.

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PANDAS 2.0 (3/4)

What's new in 2.0.0 (April 3, 2023): <https://pandas.pydata.org/docs/whatsnew/v2.0.0.html>

- ✖ A new `dtype_backend` parameter for I/O operators that support creating Arrow-backed DataFrames.

```
pd.options.mode.dtype_backend = 'pyarrow' # Set it globally
pd.read_csv(my_file, dtype_backend='pyarrow') # Set it on each DataFrame
```

- ✖ Represent “missing values” and use better support for data types outside of numerical types
 - + NumPy has poor support for non-numerical data and a lack of missing values.
 - + Before pandas 2.0, there are different types of missing values; `np.nan` is for floating-point numbers; `None` and `np.nan` are for object types, and `pd.NaT` is for date-related types.
 - + In Pandas 1.0, `pd.NA` was introduced to avoid type conversion, but it needs to be specified manually by the user.
 - + The new `string[pyarrow]` column type is around 3.5 times more efficient.

```
pd.Series([1,2,3,4], dtype='int64[pyarrow]')
pd.Series(['foo', 'bar', 'foobar'], dtype='string[pyarrow]')
```

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PANDAS 2.0 (4/4)

What's new in 2.0.0 (April 3, 2023): <https://pandas.pydata.org/docs/whatsnew/v2.0.0.html>

```
In [1]: df2 = pd.DataFrame({'a':[1,2,3, None]}, dtype='int64[pyarrow]')
```

```
In [2]: df2.dtypes
Out[2]:
a    int64[pyarrow]
dtype: object
```

```
In [3]: df2
```

```
Out[3]:
   a
0   1
1   2
2   3
3 <NA>
```

Credit:
https://www.reddit.com/r/Python/comments/12b7w3y/everything_you_need_to_know_about_pandas_200/

```
In [1]: import pandas as pd
```

```
In [2]: pd.__version__
Out[2]: '2.0.0'
```

```
In [3]: df = pd.read_csv('pd_test.csv')
```

```
In [4]: df.dtypes
```

```
Out[4]:
name      object
address   object
number    int64
dtype: object
```

```
In [5]: df.memory_usage(deep=True).sum()
Out[5]: 17898876
```

```
In [6]: df_arrow = pd.read_csv('pd_test.csv', dtype_backend="pyarrow", engine="pyarrow")
```

```
In [7]: df_arrow.dtypes
```

```
Out[7]:
name      string[pyarrow]
address   string[pyarrow]
number    int64[pyarrow]
dtype: object
```

```
In [8]: df_arrow.memory_usage(deep=True).sum()
Out[8]: 7298876
```

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PANDAS

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SERIES AND DATAFRAME

✖ pandas.Series is a 1D array holding data of any type.

✖ pandas.DataFrame is a 2D array consisting of one or more pandas.Series (= column).

Series 1

Mango

0	4
1	5
2	6
3	3
4	1

+

Series 2

Apple

0	5
1	4
2	3
3	0
4	2

+

Series 3

Banana

0	2
1	3
2	5
3	2
4	7

=

DataFrame

	Mango	Apple	Banana
0	4	5	2
1	5	4	3
2	6	3	5
3	3	0	2
4	1	2	7

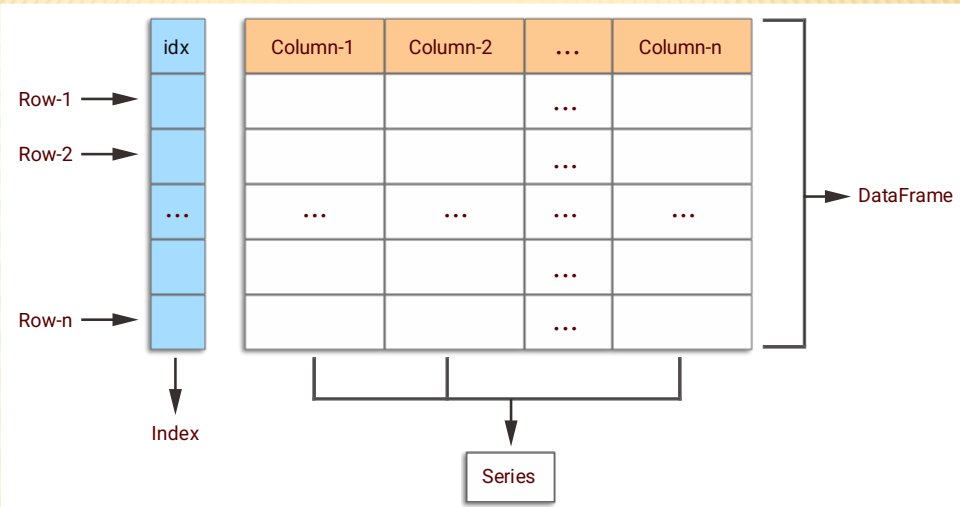
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SERIES AND DATAFRAME

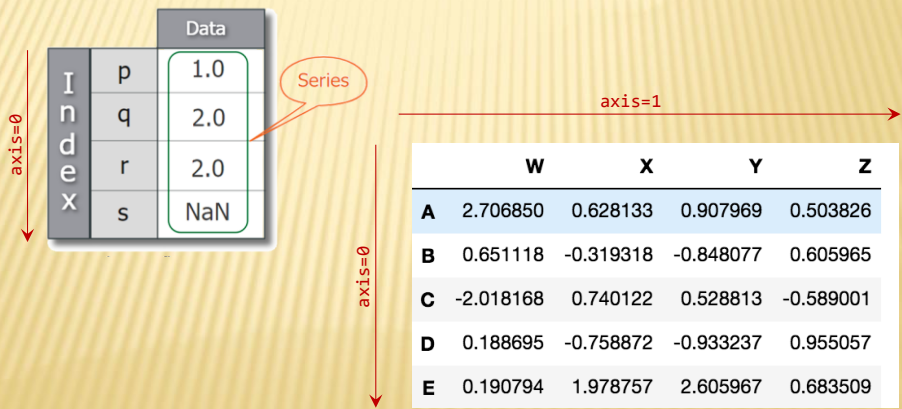


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AXIS

- ✖ pandas.Series is 1D so there is only one axis.
- ✖ pandas.DataFrame is 2D so there are two axes.



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LET'S CONTINUE IN colab

- ✖ Pandas1.ipynb
 - + IPython.display: display(), Markdown(), Latex(), Code(), HTML(), JSON(), clear_output(), ...
 - + Data types: pandas.Series, pandas.DataFrame
 - + Load, save, and render: read_csv(), to_csv(), read_excel(), to_excel(), to_html(), to_latex(), to_json(), ...
 - + Inspect, (unconditional) access, change, sort, and save data: loc[], iloc[], at[], iat[], sort_index(), sort_values(), ...
- ✖ Pandas2.ipynb
 - + Conditionally filter data with boolean indexing, query(), filter(), select_dtypes()
 - + Handle missing data and duplicated data: isna(), notna(), dropna(), fillna(), duplicated(), drop_duplicates()
- ✖ Pandas3.ipynb
 - + Aggregate data: agg(), sum(), mean(), min(), max(), median(), mode(), std(), unique(), nunique(), count(), value_count(), ...
 - + Transform data: apply(), transform()
 - + Group data: groupby(), filter()
- ✖ Pandas4.ipynb
 - + Combine data: merge(), join(), compare(), concat()
 - + Reshape data: pivot(), pivot_table(), melt()
 - + Table styling: pandas.DataFrame.style, style.format(), style.applymap(), style.apply(), style.bar(), style.highlight_max(), style.highlight_min(), ...

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LET'S CONTINUE IN colab

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A quick note on the Colab files:

- **Not Slides:** They aren't meant to be read as a lecture summary.
- **Reference Guide:** They contain detailed, multi-approach code examples for your reference.
- **Focus on Class:** In-class lectures are where we cover the core concepts and key takeaways.

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END OF THIS CLASS

- ✕ News and announcement:
 - + Microsoft Teams > General channel
- ✕ Class schedule:
 - + Microsoft Teams > General channel > Shared (Files) tab > Class Materials folder > CLASS_SCHEDULE.docx

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